ECE 447

Lesson 14

GR1 Review Day



Schedule and Admin

- Schedule.
  - Lesson 14 - **GR1 Review Day** (today)
  - Lesson 15 - **GR1** (Thursday, 17 Sep) – covering Lessons 1-13, Chapters 1-4
  - Lesson 16 - SDR Lab 4
  - Lesson 17 - GR review and feedback
- Admin
  - **HW3**. Due tonight (Lesson 14, 15 Sep) to Gradescope – angle modulation, 45 points.
  - **Lab 3**. Due Lesson 16 (Tue, 22 Sep).

GR1 Logistics

- **When:** Lesson 15, one class period (53 minutes).
- **Coverage:** Lessons 1-13, Chapters 1-4. **Not** tested: superheterodyne receivers or image frequencies – but frequency mixing and FM demodulation still fair game.
- **Bring:** calculator and a pencil. You get the **course reference sheet**: equations on page 1, Fourier transform pairs and properties on page 2. **No handwritten notes page!**
- **Show your work.** Partial credit is only possible if I can follow what you did. Be neat and box your answers.

Today is a **working session**. Five problems, none of them repeats of a homework problem. Solutions will be posted after class.

Recap: Signals & Spectral Density (Ch. 2-3)

- **Size of a signal.** Energy $E_g = \int_{-\infty}^{\infty} |g(t)|^2\,dt$; power $P_g = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} |g(t)|^2\,dt$. Energy signals have $E_g < \infty$, $P_g = 0$; power signals the reverse. A sinusoid is a power signal, a decaying exponential is an energy signal.
- **Normalized resistance.** $R = 1\,\Omega$, so $p(t) = |x(t)|^2$ and answers come out in W and J directly.
- **Parseval / Rayleigh.** $E_g = \int |g(t)|^2\,dt = \int |G(f)|^2\,df$. The integrand on the right is the **ESD** $\Psi_g(f) = |G(f)|^2$.
- **Essential bandwidth.** Solve $\int_{-B}^{B} \Psi_g(f)\,df = \alpha E_g$ for B, where α is the decimal value of the the desired percent energy. Remember to use symmetry to simplify problems.
- **Autocorrelation.** $\psi_g(\tau) = \int g(t)g^*(t-\tau)dt \leftrightarrow \Psi_g(f)$. Two routes to the same ESD; use whichever is less work.
- **LTI systems.** $Y(f) = H(f)X(f)$. **Distortionless** transmission needs $|H(f)|$ *constant* and $\angle H(f)$ *linear* across the signal band, giving $y(t) = kx(t - t_d)$ with $t_d = -\frac{1}{2\pi}\frac{d\theta}{df}$. Violate the first and you get amplitude distortion; the second, phase (delay) distortion.

Recap: Amplitude Modulation (Ch. 4.1-4.4)

- **DSB-SC.** $\varphi(t) = m(t)\cos\omega_c t$. Spectrum is $M(f)$ shifted to $\pm f_c$ and halved. Transmission bandwidth $2B$. Power $P = \frac{1}{2}P_m$. Needs a **coherent** detector – a local oscillator phase error θ scales the output by $\cos\theta$, and $\theta = 90^\circ$ kills it entirely.
- **DSB-LC ("AM").** $\varphi(t) = [A + m(t)]\cos\omega_c t$. Modulation index $\mu = \frac{m_p}{A}$. Envelope detection works only if $\mu \leqslant 1$; $\mu > 1$ is overmodulation and the envelope no longer follows $m(t)$.
- **AM power.** $P_c = \frac{A^2}{2}$, $P_s = \frac{P_m}{2}$, and efficiency $\eta = \frac{P_s}{P_c + P_s}$. For a single tone this reduces to $\eta = \frac{\mu^2}{\mu^2+2}$, which maxes out at 33.3% when $\mu = 1$ – the carrier carries no information but most of the power.
- **Bandwidth-efficient cousins.** SSB halves the bandwidth to B; VSB is the practical compromise; QAM sends two messages in the same band using quadrature carriers, and depends entirely on carrier phase accuracy.

Recap: Angle Modulation (Ch. 4.5-4.9)

- **The generalized carrier.** $\varphi(t) = A_c \cos\theta(t)$ with $\omega_i(t) = \frac{d\theta}{dt}$. **PM:** $\theta = \omega_c t + k_p m(t)$. **FM:** $\theta = \omega_c t + k_f \int m(\alpha) d\alpha$. Power is always $\frac{A_c^2}{2}$ – it does not depend on the message, which is the whole point of angle modulation.
- **Deviation.** $\Delta f_{FM} = \frac{k_f m_p}{2\pi}$ and $\Delta f_{PM} = \frac{k_p \dot{m}_p}{2\pi}$. Note what each depends on: FM deviation tracks the message *amplitude* only, PM deviation tracks the *slope*. Doubling the message frequency changes B_{PM} but leaves Δf_{FM} alone.
- **Carson's rule.** $B_{FM} = 2(\Delta f + B) = 2B(\beta + 1)$ where $\beta = \frac{\Delta f}{B}$. Narrowband is $\beta \ll 1$ ($B_{FM} \approx 2B$, like AM); wideband is $\beta \gg 1$ ($B_{FM} \approx 2\Delta f$).
- **Demodulation.** Differentiate then envelope-detect.

Problem 1: Energy, ESD, and Bandwidth

Consider the energy signal, measured across a normalized $1\,\Omega$ resistance:

$$g(t) = 10e^{-100t}u(t) \text{ V}$$

- **(a)** Find the total energy E_g directly in the time domain.
- **(b)** Find $G(f)$ and the energy spectral density $\Psi_g(f)$.
- **(c)** Verify your answer to (a) using Parseval's theorem.
- **(d)** Find the essential bandwidth B containing 90% of E_g, using the one-sided definition (0 to B).

Useful: $\displaystyle\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a}$

Problem 2: Distortionless Transmission

A channel has the transfer function

$$H(f) = \begin{cases} 0.5\, e^{-j2\pi f t_d}, & |f| \leqslant 8\ \text{kHz} \\ 0, & \text{otherwise} \end{cases} \qquad t_d = 2\ \text{ms}$$

- **(a)** A signal $x(t)$ band-limited to 5 kHz is sent through the channel. Is the transmission distortionless? Justify using both conditions.
- **(b)** Write the output $y(t)$ in terms of $x(t)$.
- **(c)** The message bandwidth is now increased to 10 kHz. What kind of distortion appears, and why?
- **(d)** A second channel has $|H_2(f)| = 0.5$ over the same band but phase $\theta_2(f) = -2\pi(2 \times 10^{-3})f + 0.4\sin(2\pi f/8000)$. Is *it* distortionless? What changes?

Problem 3: DSB-SC

A message $m(t) = 4\cos(2\pi \cdot 1000t) + 2\cos(2\pi \cdot 3000t)$ V is transmitted DSB-SC on a 50 kHz carrier, $c(t) = \cos(2\pi \cdot 50{,}000t)$.

- **(a)** Write $\varphi_{DSB\text{-}SC}(t)$ as a sum of cosines. What are the four transmitted frequencies?
- **(b)** Sketch the double-sided amplitude spectrum, labeling every frequency and amplitude.
- **(c)** What is the transmission bandwidth?
- **(d)** Find the transmitted power, and confirm it equals $\frac{1}{2}P_m$.
- **(e)** The receiver's local oscillator drifts to a phase error of 60°. By what factor does the recovered message amplitude drop? What happens at 90°?

Problem 4: DSB-LC

An AM transmitter produces, across $R = 1\,\Omega$:

$$\varphi_{AM}(t) = [5 + 3\cos(2\pi \cdot 2000t)]\cos(2\pi \cdot 200{,}000t) \ \text{V}$$

- **(a)** Find the modulation index μ.
- **(b)** Find the carrier power P_c and the total sideband power P_s.
- **(c)** Find the power efficiency η. Confirm it against $\eta = \mu^2/(2 + \mu^2)$.
- **(d)** What carrier amplitude A would give $\mu = 1$, and what is η there?
- **(e)** The carrier amplitude drifts down to $A = 2$. Sketch the envelope and explain what a diode envelope detector would now recover.

Problem 5: Angle Modulation Design

A transmitter produces $\varphi(t) = 20\cos\big(2\pi \times 10^7 t + \theta_d(t)\big)$ V from the message $m(t) = 2\cos(2\pi \cdot 5000t)$ V.

- **(a)** For **FM** with $k_f = 4000\pi$ rad/s per volt, write the phase deviation $\theta_d(t)$ and find Δf and β.
- **(b)** Estimate the transmission bandwidth using Carson's rule.
- **(c)** The link budget requires a **narrowband** signal, $\beta \leqslant 0.2$. Holding the message fixed, what is the largest usable k_f?
- **(d)** You instead build this as **PM**. What k_p produces the same Δf as part (a)?
- **(e)** Sketch the instantaneous frequency $f_i(t)$ for the FM signal of part (a), labeling its maximum, minimum, and rate of variation.